AL RYADA UNIVERSITY FOR SCIENCE AND TECHNOLOGY

FACULTY OF NURSING

Lecture 2

Bacterial Morphology

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Lecture Learning Outcomes

By the end of this lecture, the student will be able to:

- ✓ **Define** the basic **morphology** of bacteria.
- ✓ **Describe** the general structure of the bacterial cell.
- ✓ **Identify** the different **shapes** of bacteria (cocci, bacilli, spirilla).
- ✓ **List** the main components of the bacterial cell.
- ✓ **Explain** the **functions** of each bacterial cell structure (cell wall, cell membrane, capsule, flagella, pili).
- ✓ **Differentiate** between **Gram-positive** and **Gram-negative** bacterial cell walls.
- ✓ **Recognize** the importance of bacterial structures in **pathogenicity** and **infection.**

Bacteria

- **A prokaryotic** and **single-cell** microorganism.
- **Small** in size.
- Do not have a **nuclear membrane** and also lack **membrane-bound organelles**.
- Can be seen using a **microscope.**

Bacterial morphology

Bacterial morphology

- They are prokaryotic cells.
- Bacterium is a single cell that can replicate by simple division called binary fission.

S : Size.

S :Shape.

S :Stain.

S :Spore.

M : Motility

A :Arrangement.

C :Capsule.

SSSSMAC

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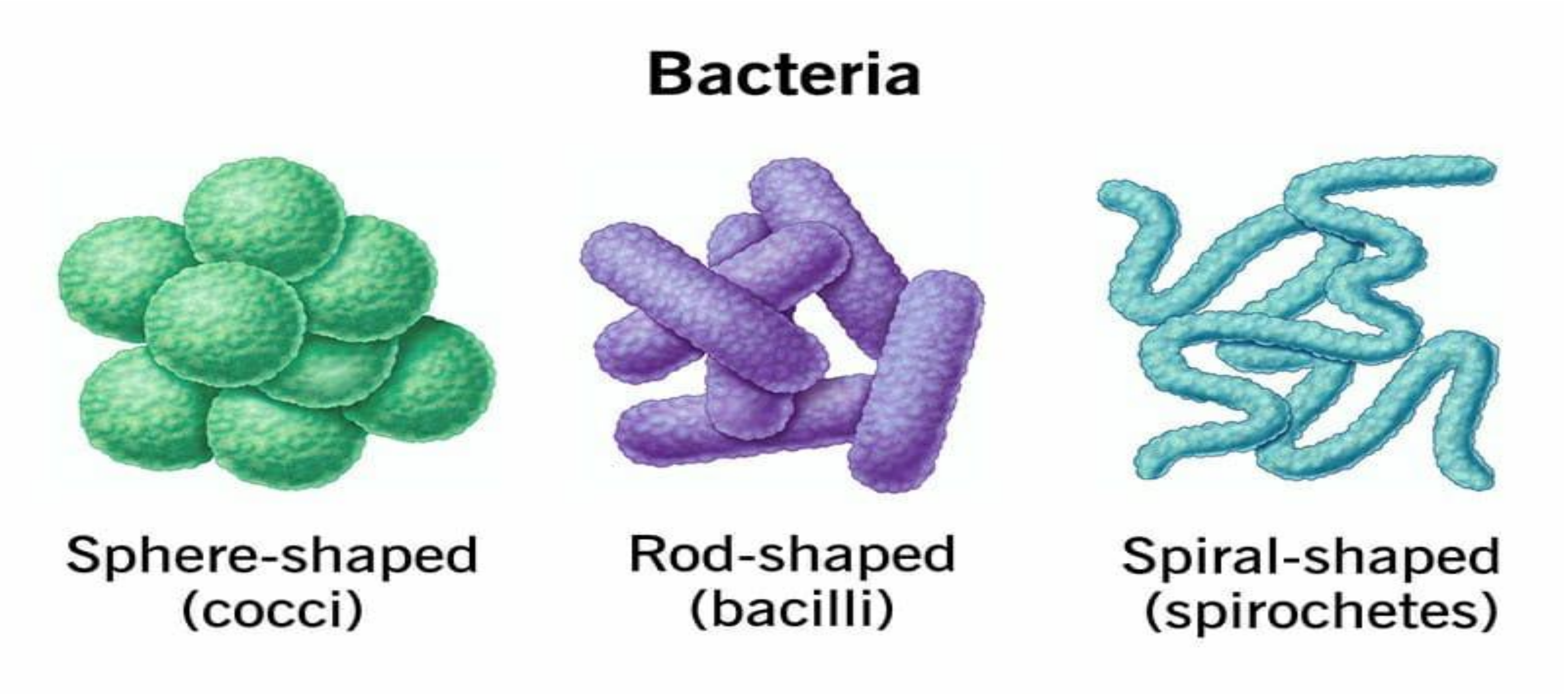
Size

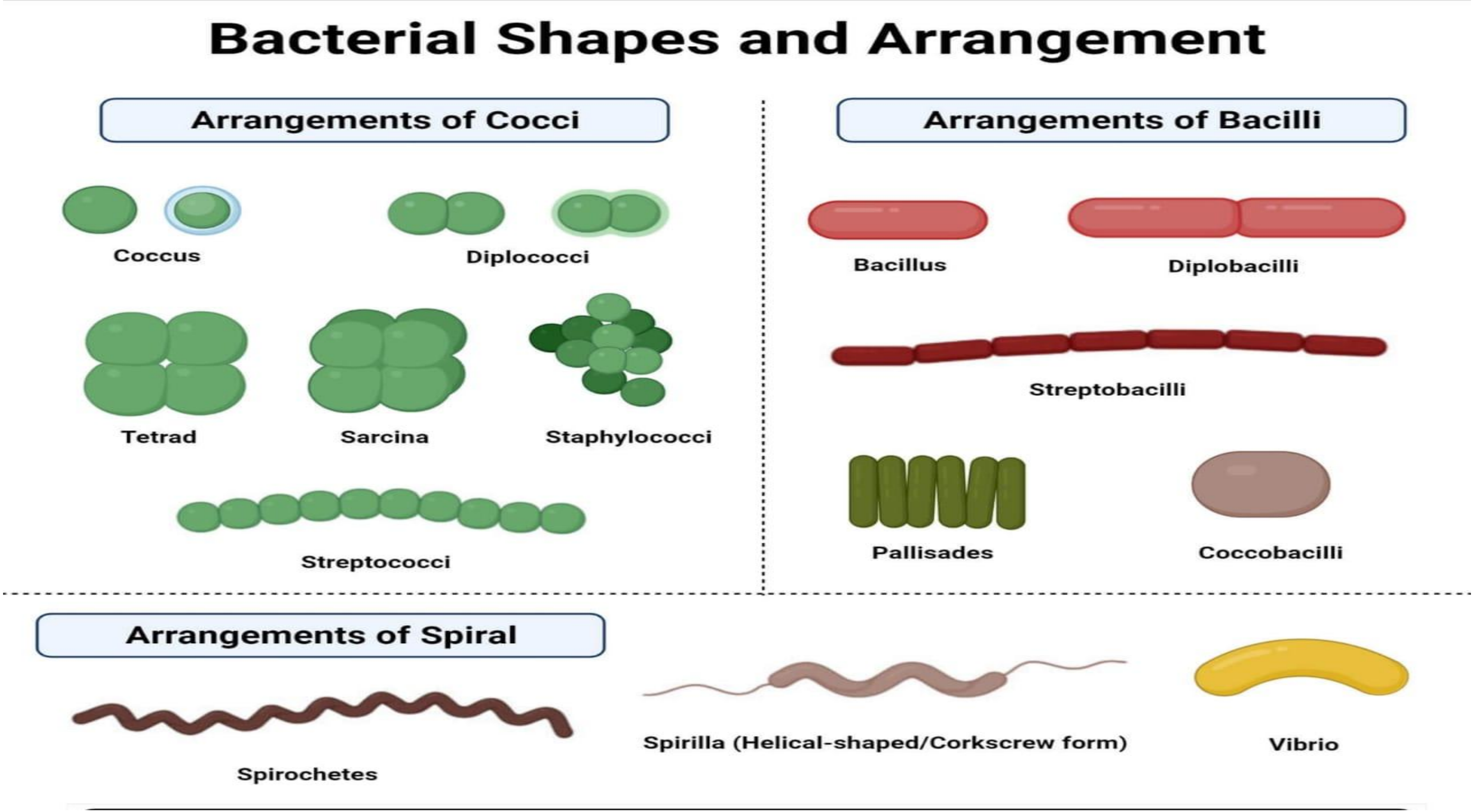
Bacteria are very small in size.

- The unit of measurement in bacteriology is the **micron** or

  micrometer (μm.). **(1 μm = 0.001 mm, 10^{-6} meter)**
- Average bacteria **0.1 - 5 um** in diameter.

Shapes

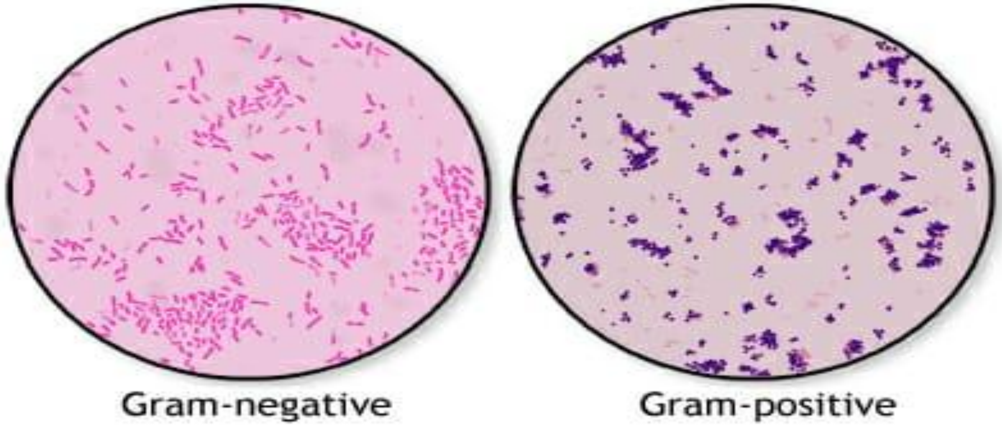

Bacteria
Sphere-shaped
(cocci)
Rod-shaped
(bacilli)
Spiral-shaped
(spirochetes)


Bacterial Shapes and Arrangement
Arrangements of Cocci
Coccus
Diplococci
Tetrad
Sarcina
Staphylococci
Streptococci
Arrangements of Bacilli
Bacillus
Diplobacilli
Streptobacilli
Pallisades
Coccobacilli
Arrangements of Spiral
Spirochetes
Spirilla (Helical-shaped/Corkscrew form)
Vibrio

Staining properties

According to Gram stain, bacteria are divided into 2 groups:

- Gram positive bacteria(violet)
- Gram negative bacteria(red or pink)



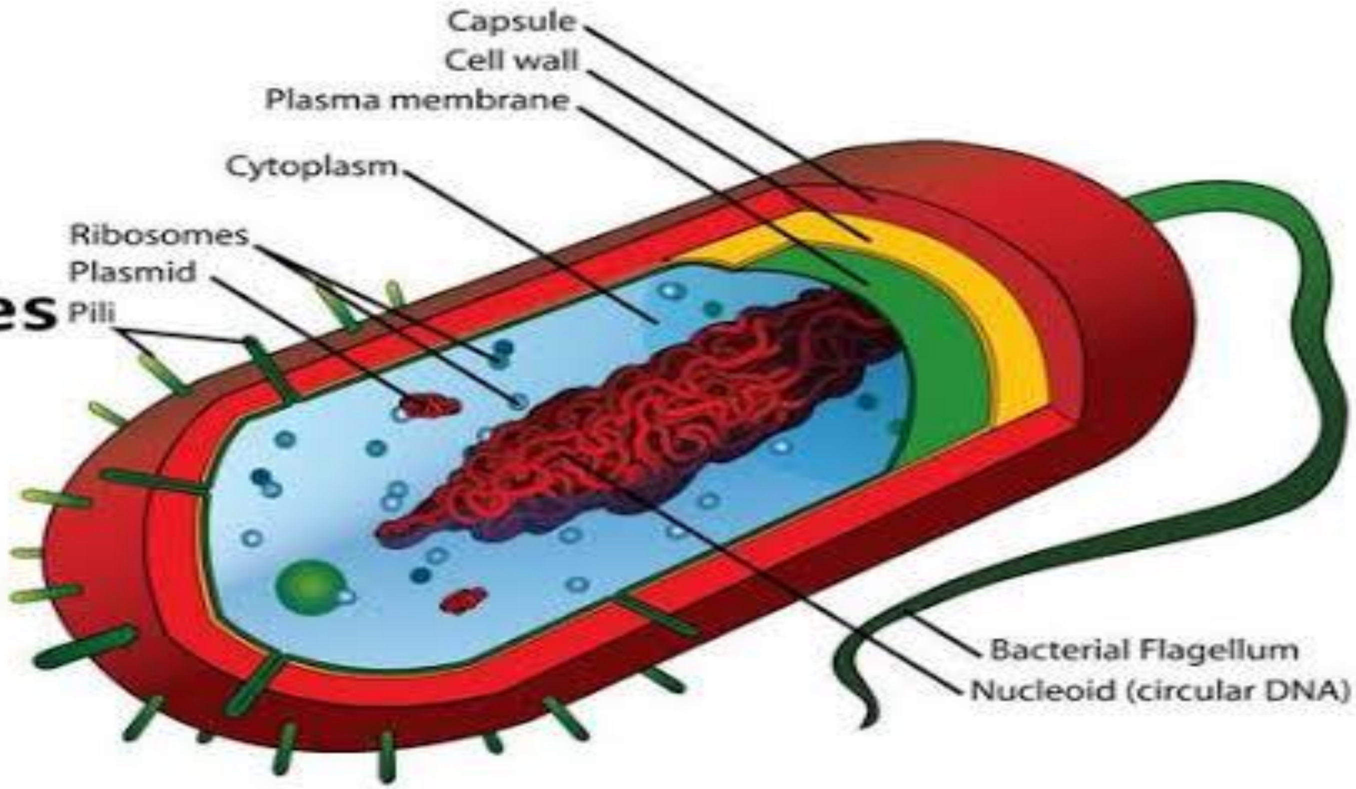
Structure of Bacteria

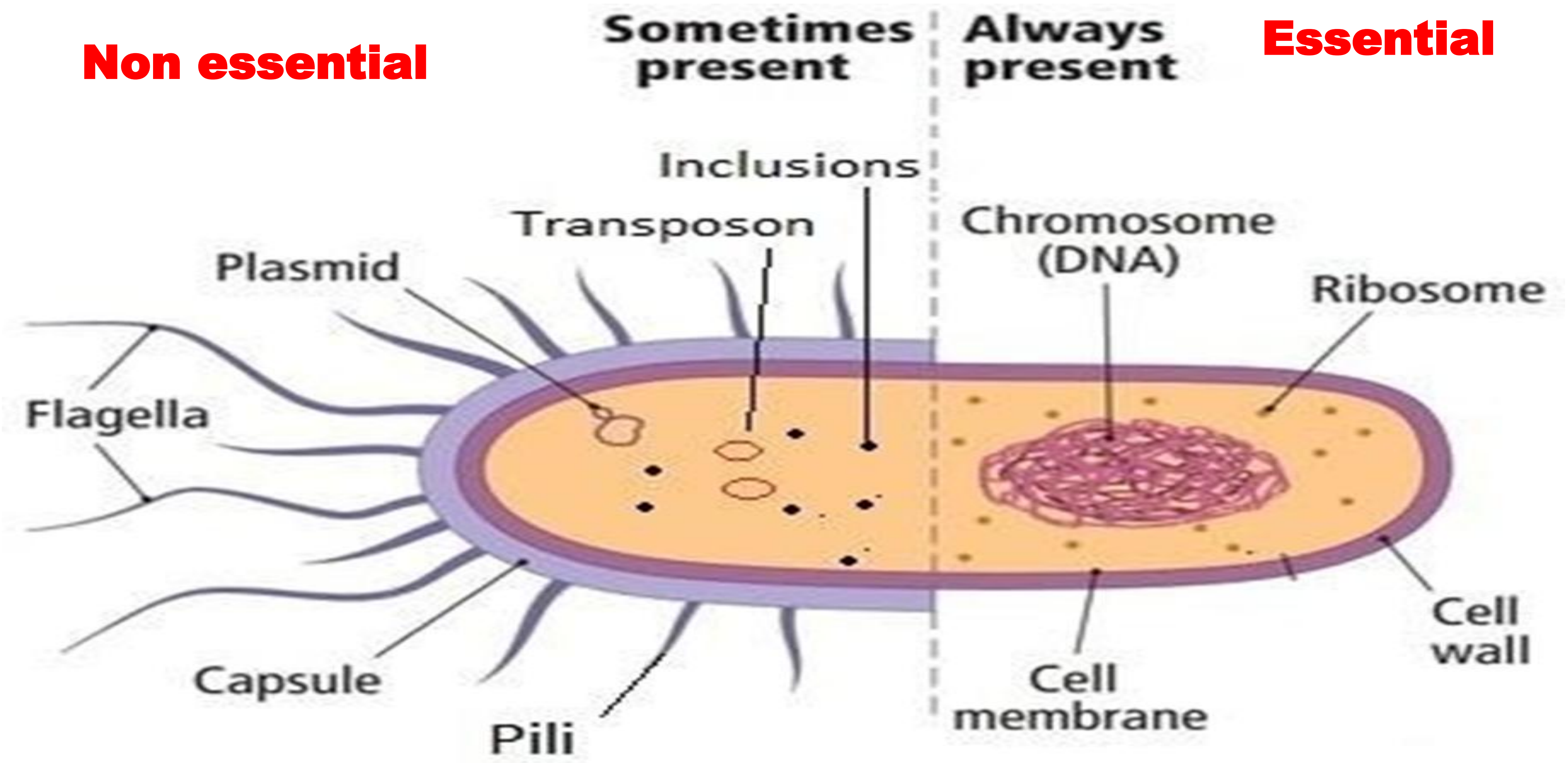
Essential structure

- Cell wall
- cell membrane
- Cytoplasm
- Nuclear material

Particular structures

- Capsule
- flagella
- pili
- Spore




Non essential
Sometimes present
Always present
Essential
Inclusions
Transposon
Plasmid
Chromosome (DNA)
Ribosome
Flagella
Capsule
Pili
Cell membrane
Cell wall

Cell wall

- It is the outermost component

- It is a multilayered structure.

- Formed mainly of **Peptido-glycan**

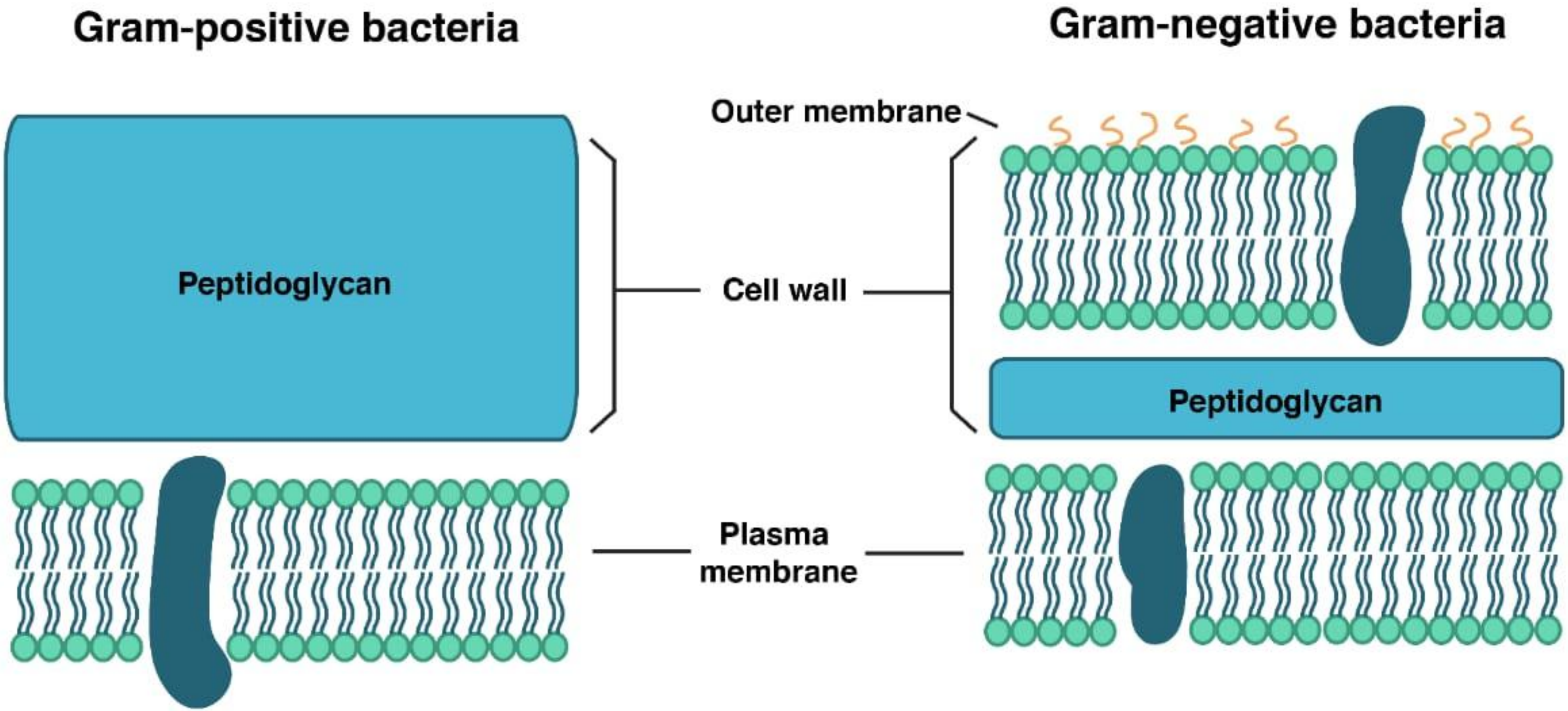
Polymer (amino acids + sugars)

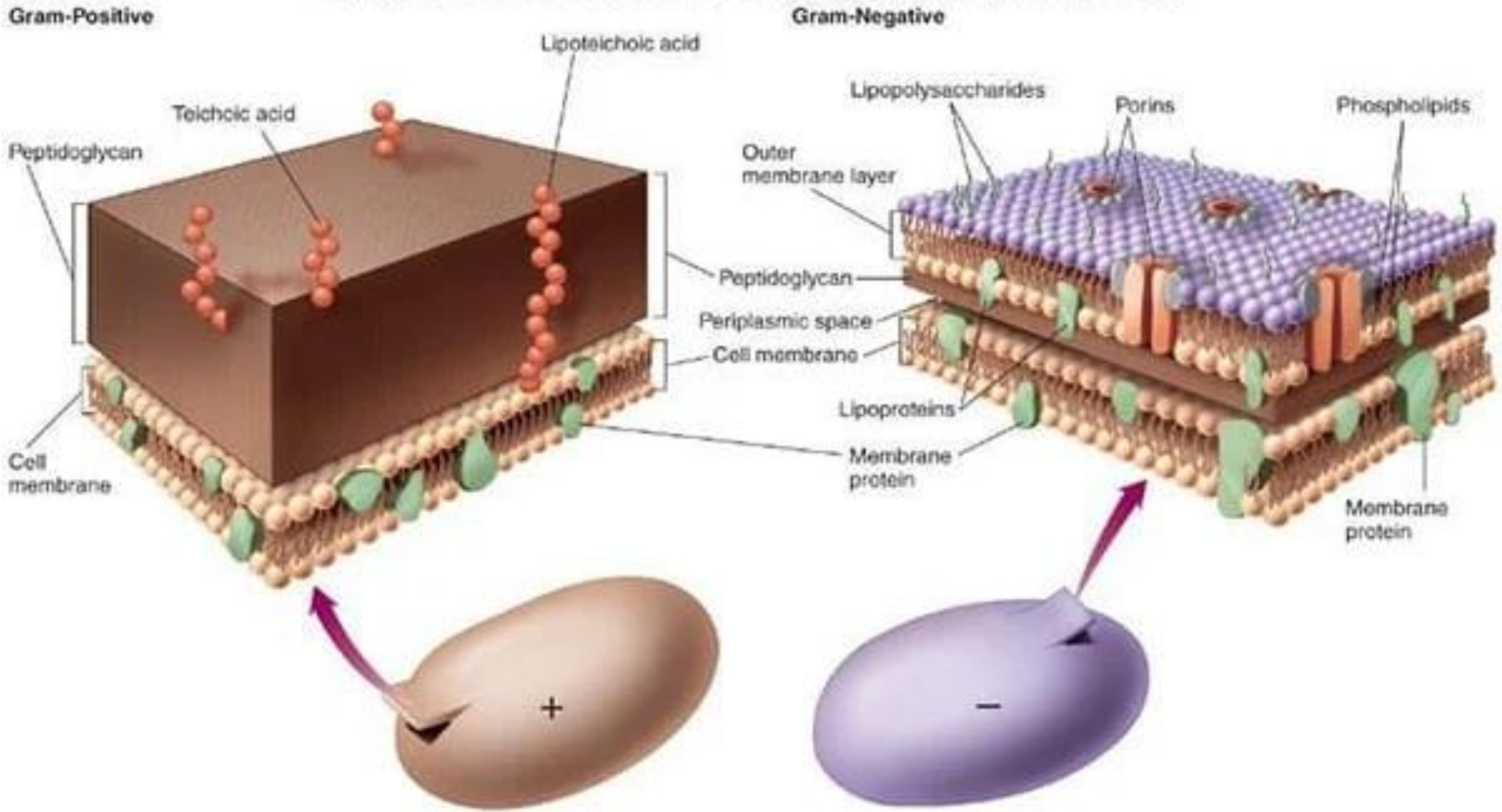
Bacteria are classified into two major groups based on the structure of their cell wall and their reaction to the Gram stain technique:
Gram-Positive bacteria
Gram-Negative bacteria
This classification depends on the **composition and thickness of the peptidoglycan** layer in the bacterial cell wall.

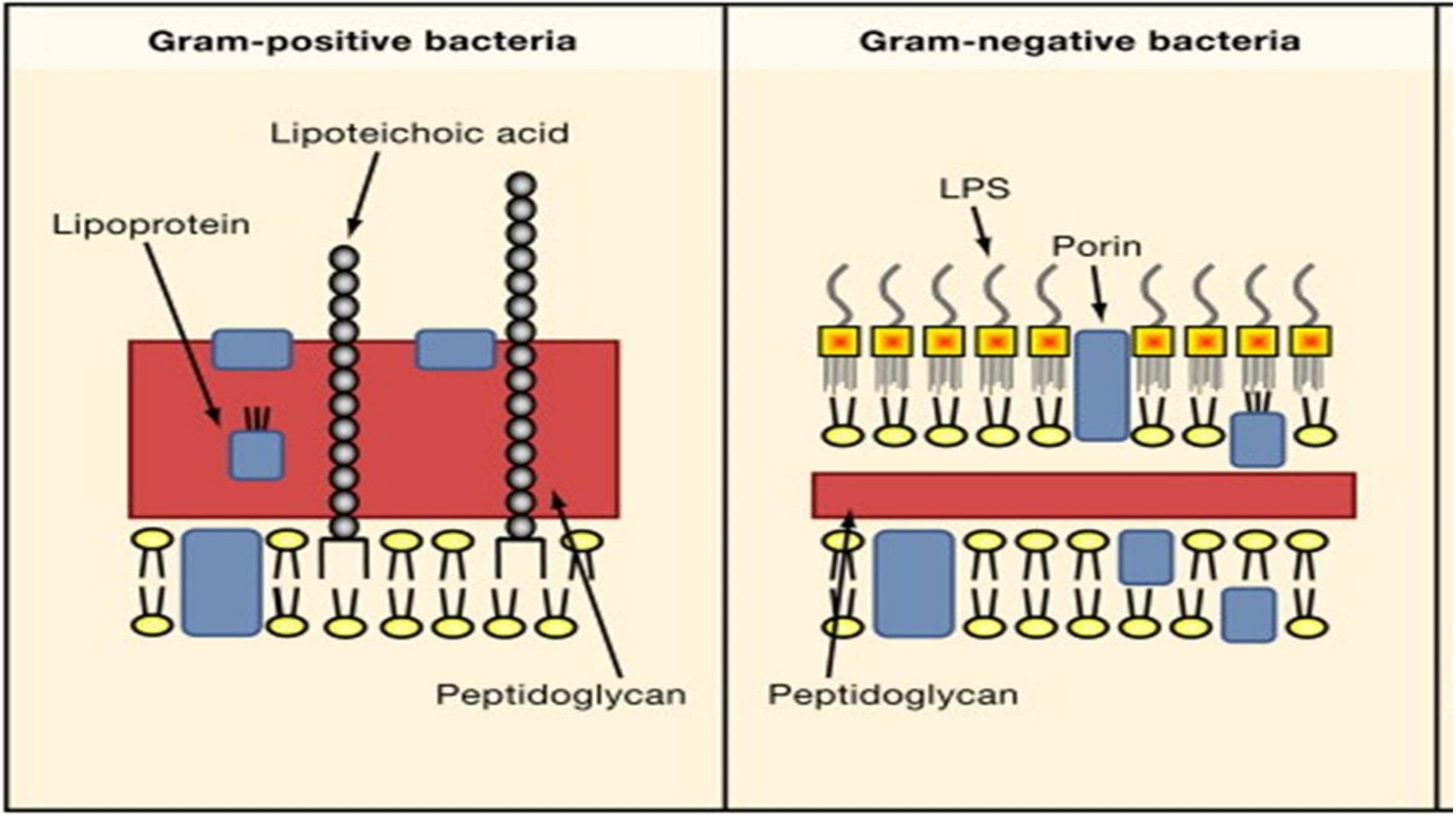
Gram-positive bacteria have a **thick peptidoglycan** layer, which allows them to retain the **crystal violet** stain and appear **purple** under the microscope.

In contrast, **Gram-negative bacteria** have a **thin peptidoglycan** layer and an outer lipid membrane, causing them to lose the crystal violet stain and take up the counterstain (**safranin),** appearing **pink or red.**

This classification plays a role in the **diagnosis** and **treatment** of bacterial **infections** in clinical practice.


Gram-positive bacteria
Gram-negative bacteria
Outer membrane
Peptidoglycan
Cell wall
Peptidoglycan
Plasma
membrane


Gram-Positive
Gram-Negative
Lipoteichoic acid
Teichoic acid
Peptidoglycan
Cell membrane
Lipopolysaccharides
Porins
Phospholipids
Outer membrane layer
Peptidoglycan
Periplasmic space
Cell membrane
Lipoproteins
Membrane protein
Membrane protein
+
−


Gram-positive bacteria
Lipoteichoic acid
Lipoprotein
Peptidoglycan
Gram-negative bacteria
LPS
Porin
Peptidoglycan

	Gram +ve bacteria	Gram –ve bacteria
Peptidoglycan	Thicker	Thinner
Teichoic acid	yes	no
Outer membrane	no	yes
Periplasmic space	no	yes
Antigens	Teichoic acids	LPS

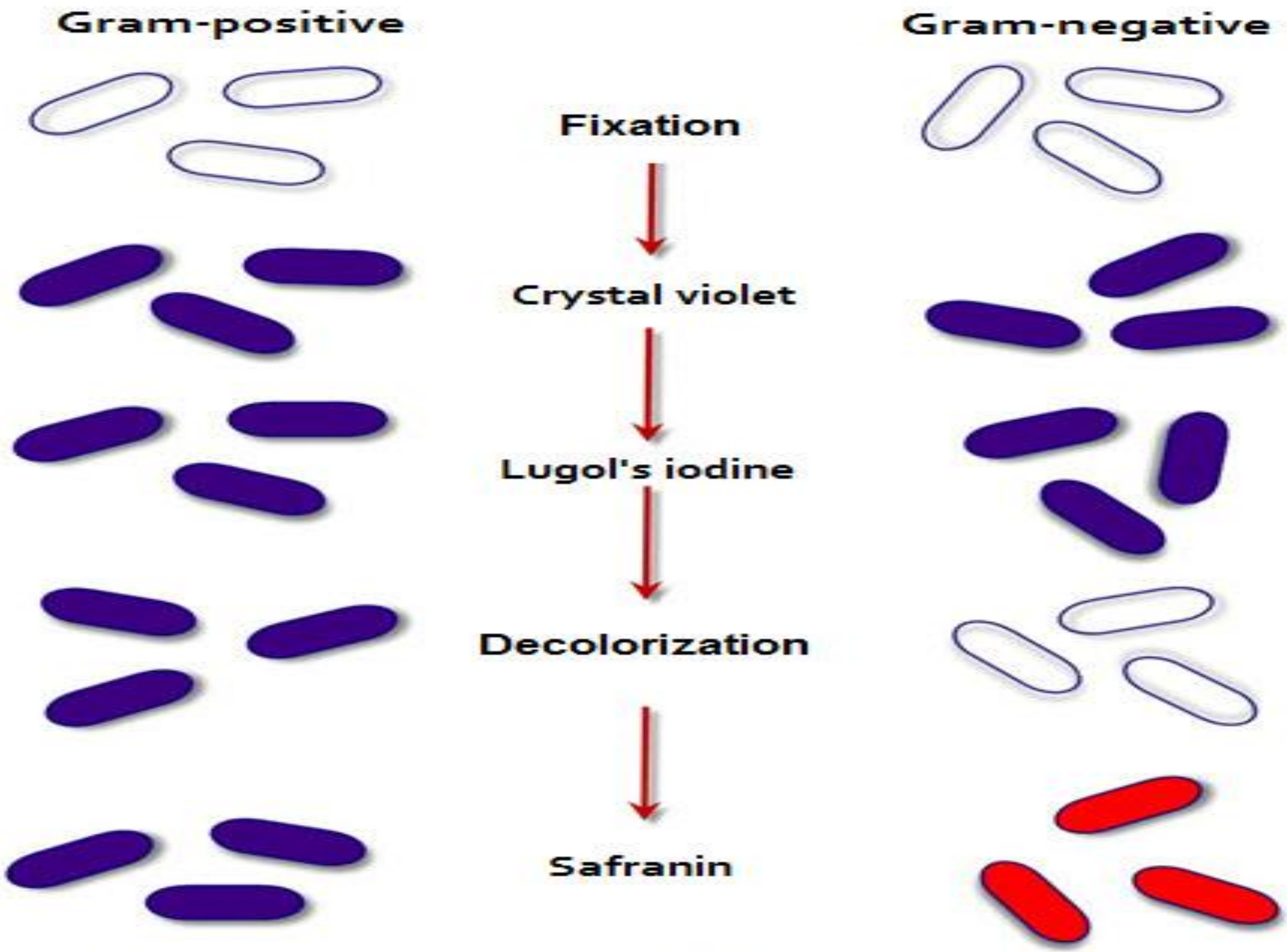


Diagram representation of the bacterial staining
staining process

Function of cell wall:

1-Rigid support.

2-Channel to allow the entry of essential substance as sugars, amino acids, vitamins and metals

3-Cell division.

4-Determining the reaction to Gram stain.

5-Antigenic substances.

6-Good target for antibacterial drugs

Cell Membrane (Cytoplasmic Membrane or Plasma Membrane)

- Just internal to the cell wall
- It is a semi-permeable thin membrane
- formed of 2 layers of phospholipids in which proteins and enzymes are embedded.

Function:

a) **Active transport of molecules** into the cell by permease enzymes.
b) Energy generation by **oxidativephosphorylation** (**mitochondrial analogue).**
c) Synthesis of **precursors of the cell wall.**
d) **Mesosomes.**

Mesosome

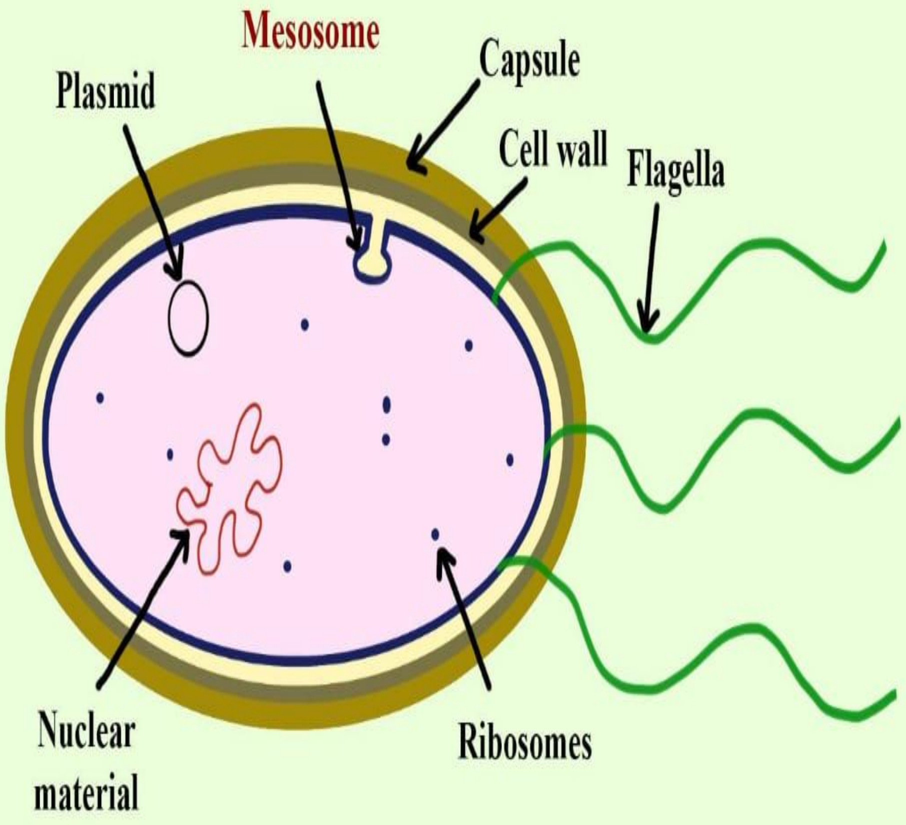
It is an invagination of the cytoplasmic membrane.

Function:

It is important during cell division as it acts as:

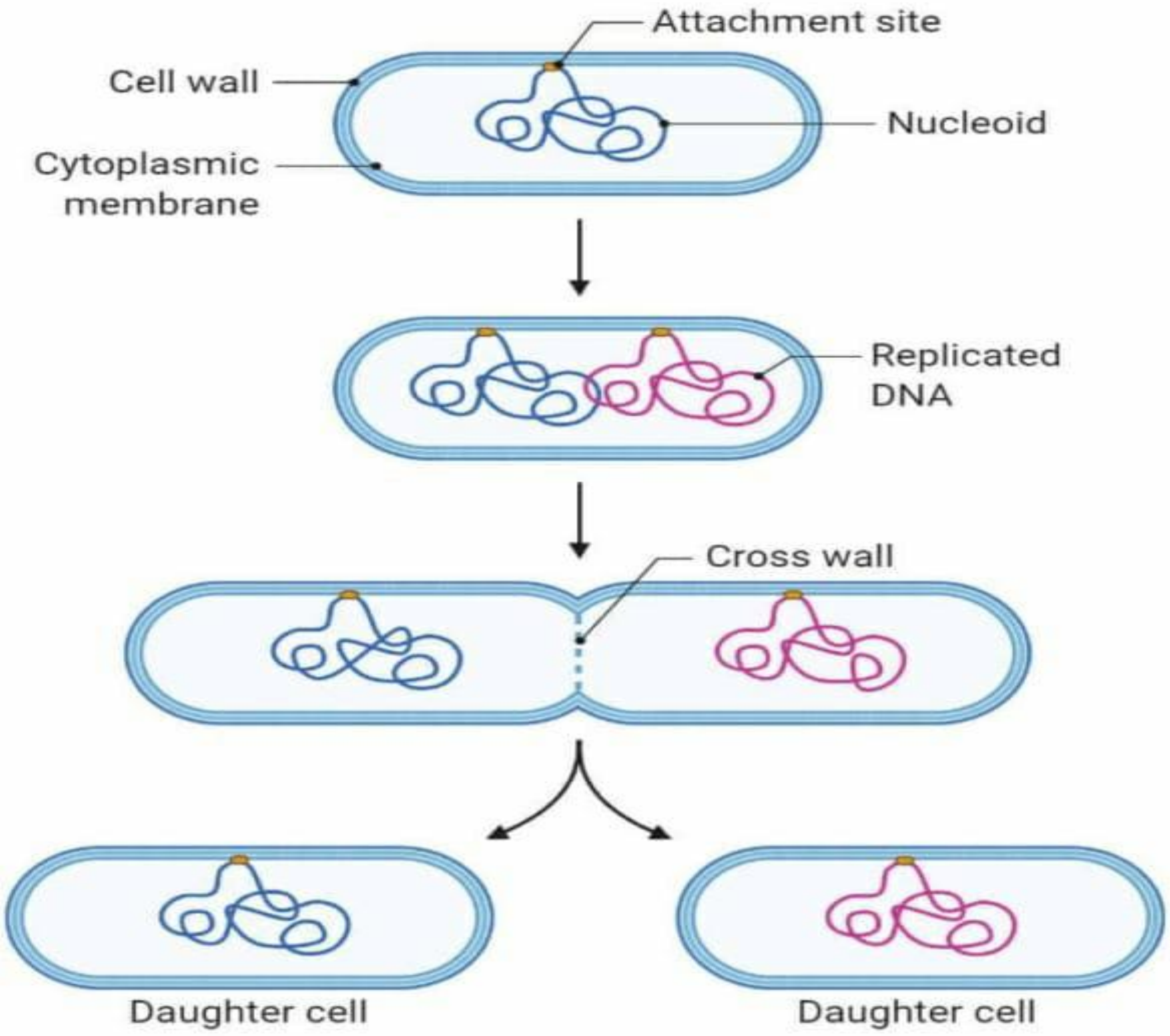
1.The origin of the transverse septum.

2.The binding site of the DNA which will become
the genetic material of each daughter cell.



Mesosome

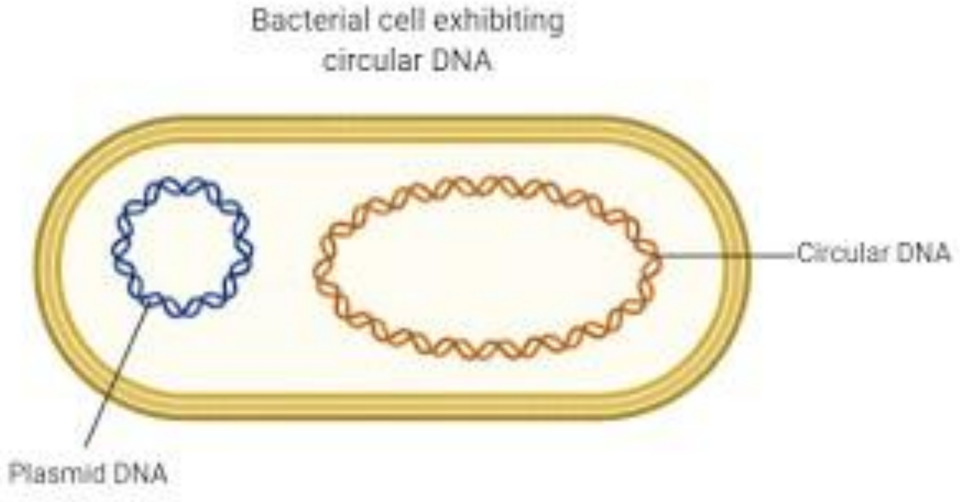
Prokaryotic Cell Division by Binary Fission



1. Prokaryotic parent cell initiates replication
2. A copy of the cell's DNA is created
3. Cell elongates and cross wall forms
4. Cross wall forms completely and daughter cells separate

Intracytoplasmic structures

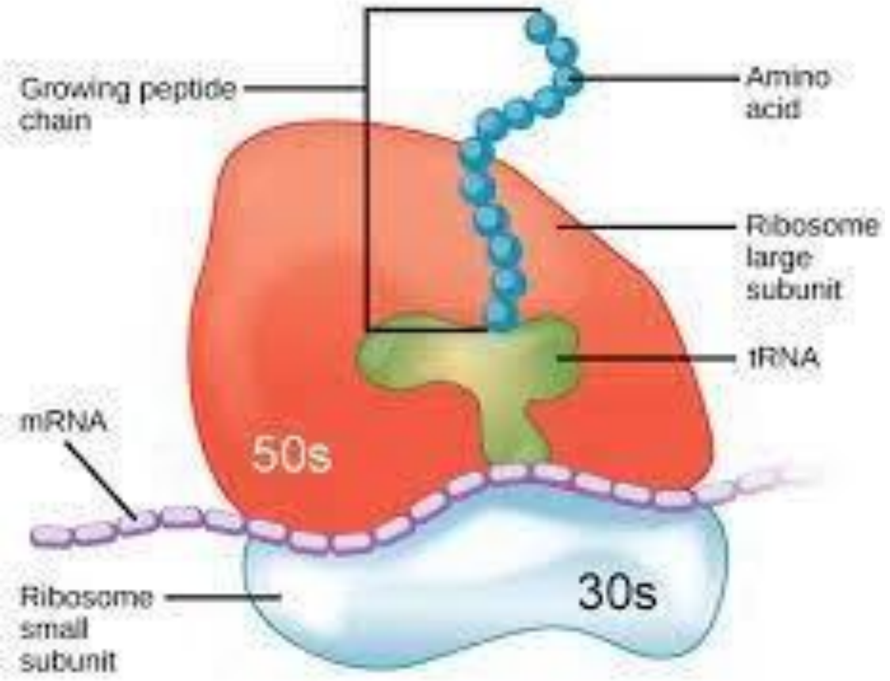
1) Nuclear material:



- Single circular double stranded DNA molecule (chromosome)
- No nuclear membrane

2) Ribosomes:

- Site of protein synthesis
- Spherical particles in the cytoplasm
- Bacterial ribosomes are 70S.



3) Intracytoplasmic inclusions:

Granules of food or energy reserve.

4)Plasmids:

- Extrachromosomal, double-stranded, circular DNA molecules
- Capable of replicating independently of the bacterial chromosome

5)Transposons:

- Non-replicating pieces of DNA
- Move readily from one site to another

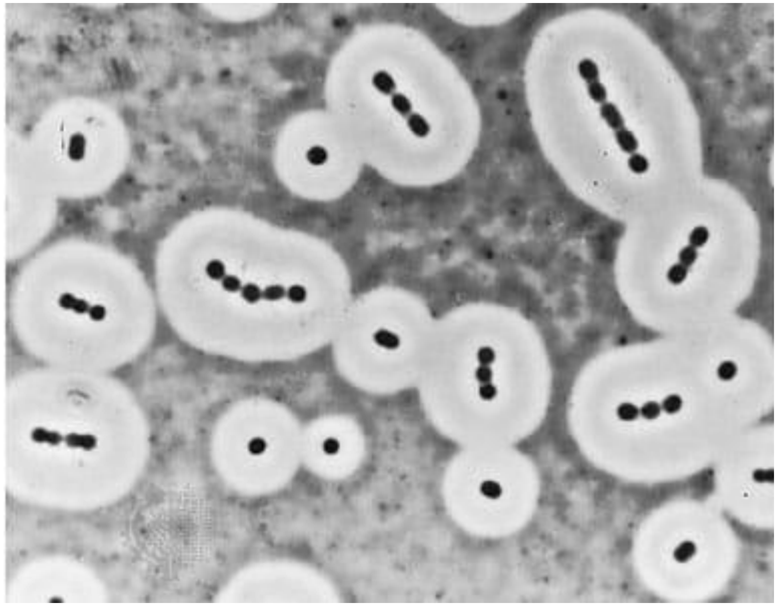
Surface structures

1)Capsule (virulence factor):

Polysaccharide gelatinous layer

Protect bacteria from phagocytosis

Function:



a)Determinant of virulence.

b)Capsular polysaccharides are used as the antigens in certain vaccines, since they are capable of stimulating protective antibodies.

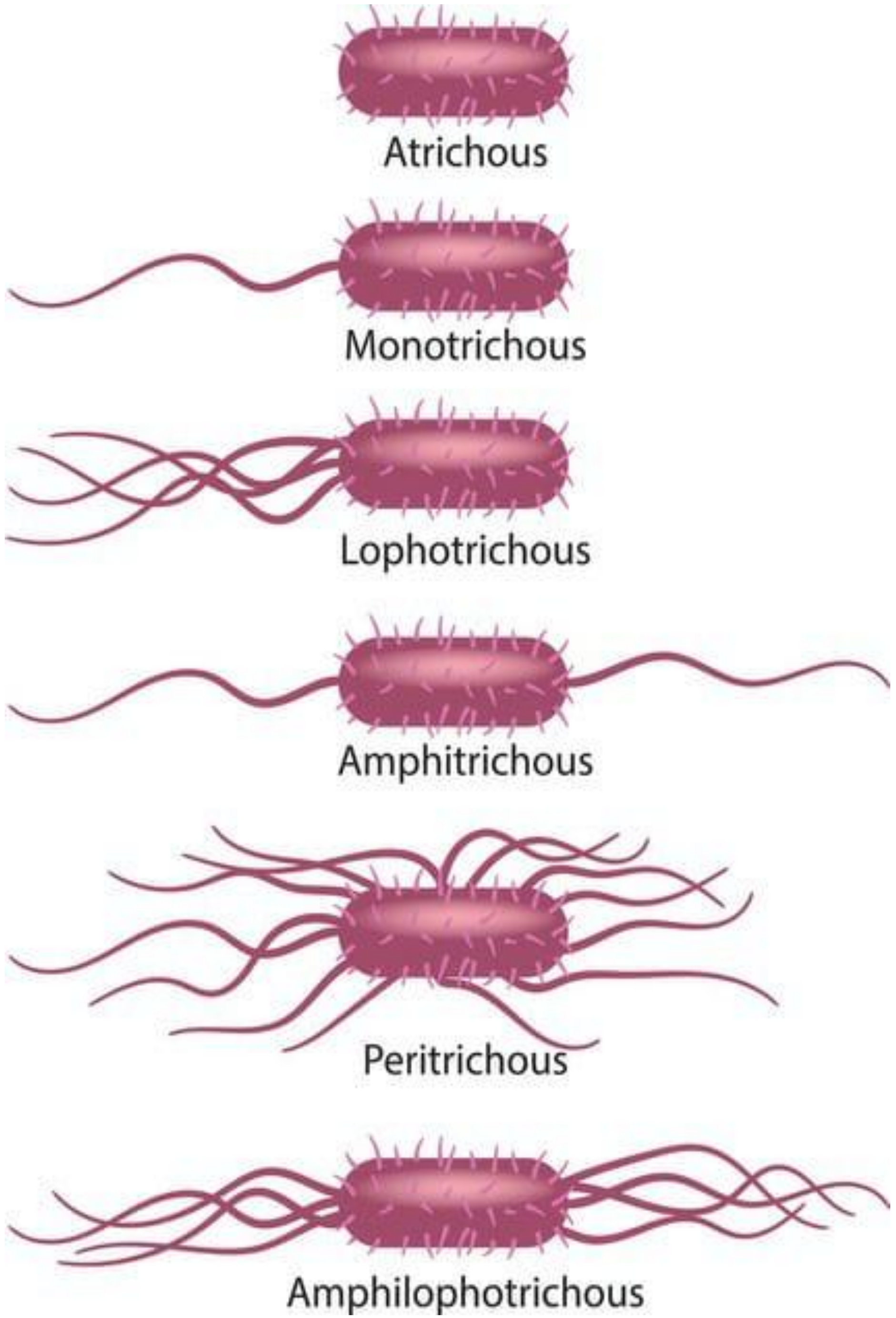
c)Identification and typing of bacteria

2)Flagella:

Organs of motility in motile bacteria.
Long, whiplike structure

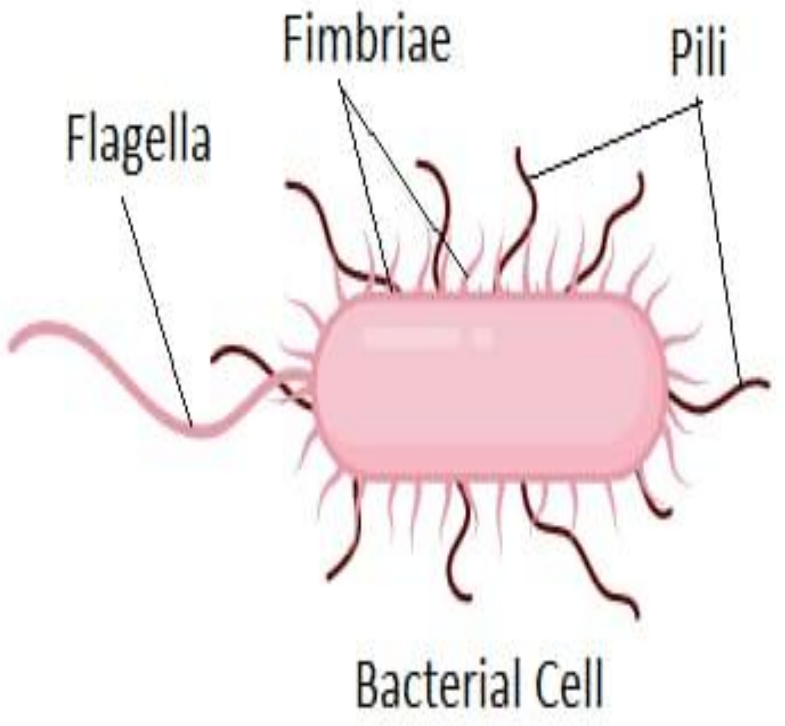
Function:

a)Move the bacteria toward nutrients.
b)Formed of protein subunits called flagellin "antigenic"
c)Typing of bacteria



3) Pili

Flagella Vs Pili



- **Hair like projections that extend from the cell membrane**
- **Shorter, thinner and straighter than flagella**
- **Protein called pilin (antigen)**
- **Mainly on Gram negative organisms**

Function:

Pili have 2 types:

a)Ordinary pili: mediate the adherence of bacteria to specific receptors on the human cell surface "virulence factor"

b)Sex pili: transfer of DNA between 2 bacteria.

Bacterial endospore

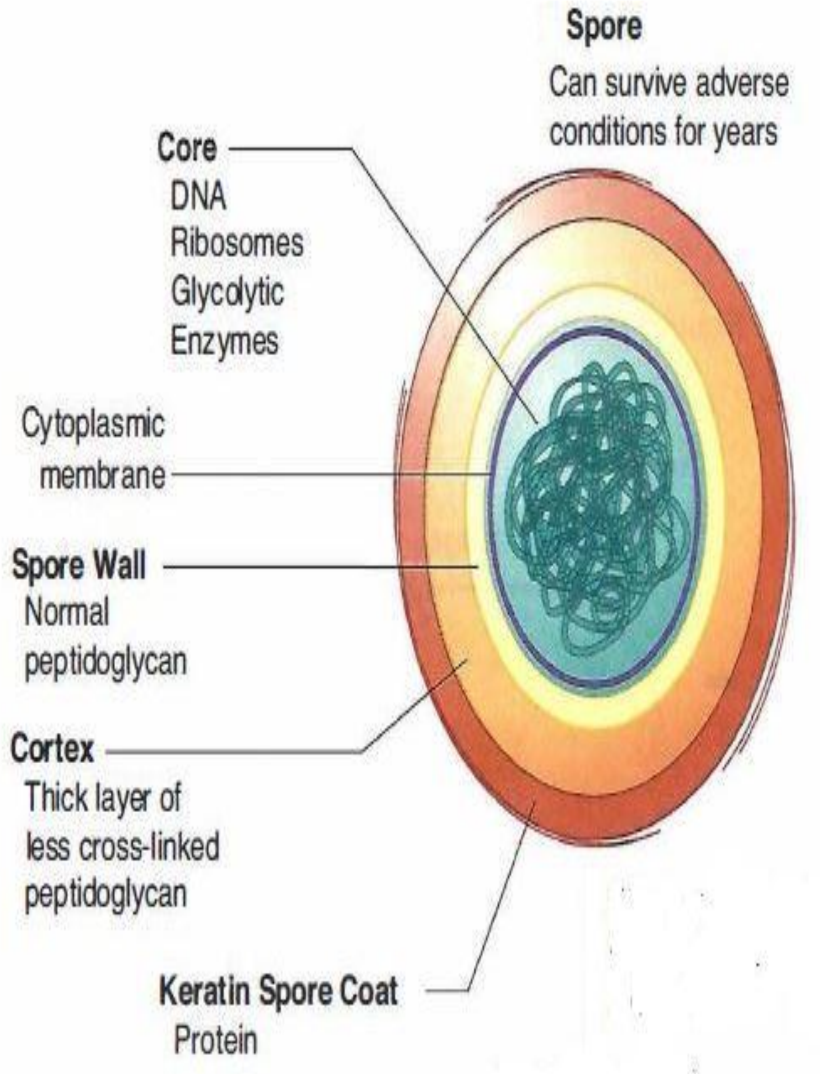
highly resistant forms of some Gram positive bacilli e.g. Clostridia.

Composition:

incorporates the bacterial DNA + a small amount of cytoplasm surrounded by very thick cortex and keratin coat.

Why?

formed in response to unfavorable conditions e.g. depletion of nutrients, heat, dryness, etc.).

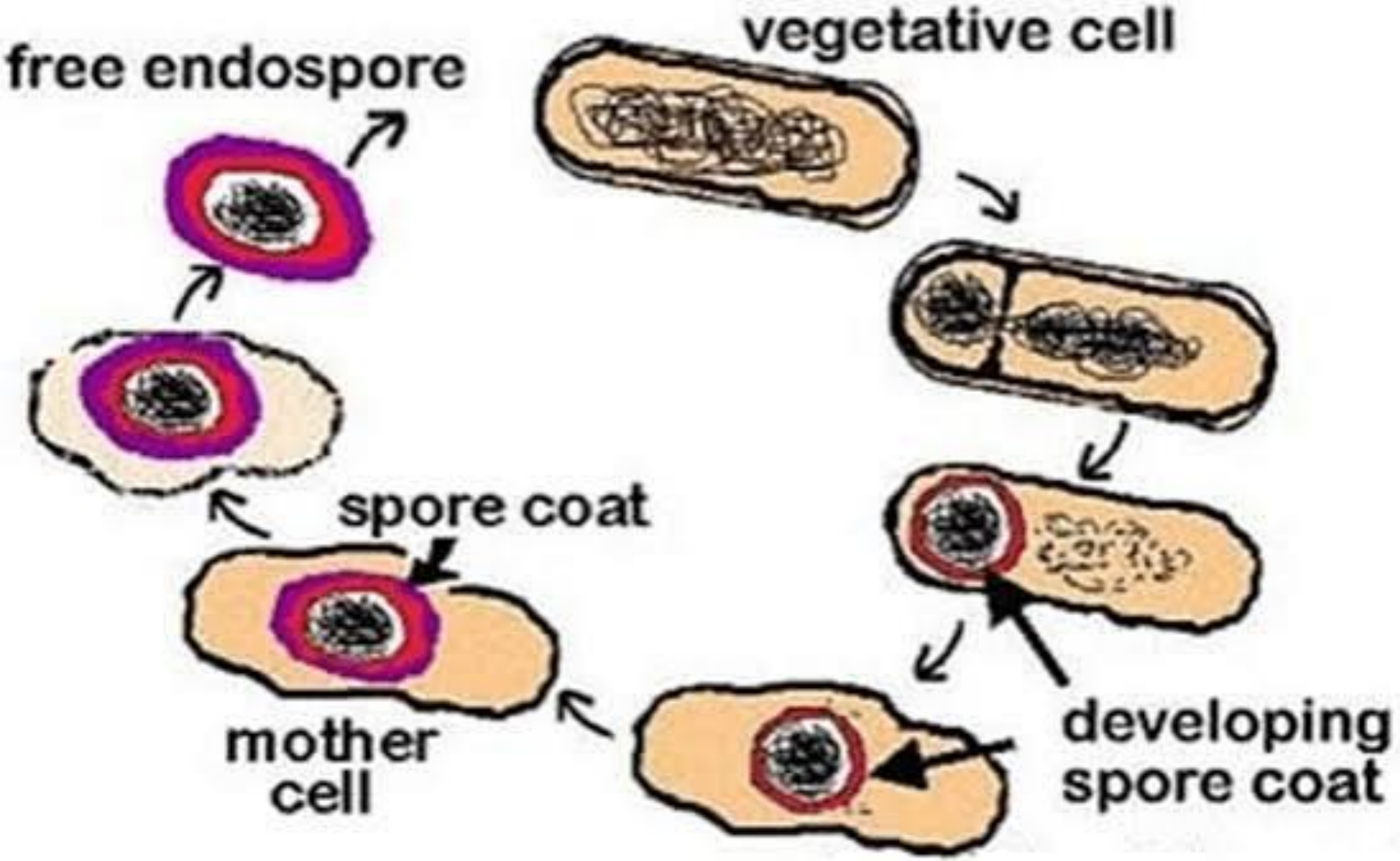


How ?

- The spore is formed inside the parent vegetative bacterial cell at the start of sporulation process so it is known as endospores

- Once formed, the spore has no metabolic activity and can remain dormant for many years.

- Upon exposure to water and the appropriate nutrients ⇨ specific enzymes degrade the coat ⇨ water and nutrients enter ⇨ and germination into a metabolizing, reproducing vegetative bacterial cell.

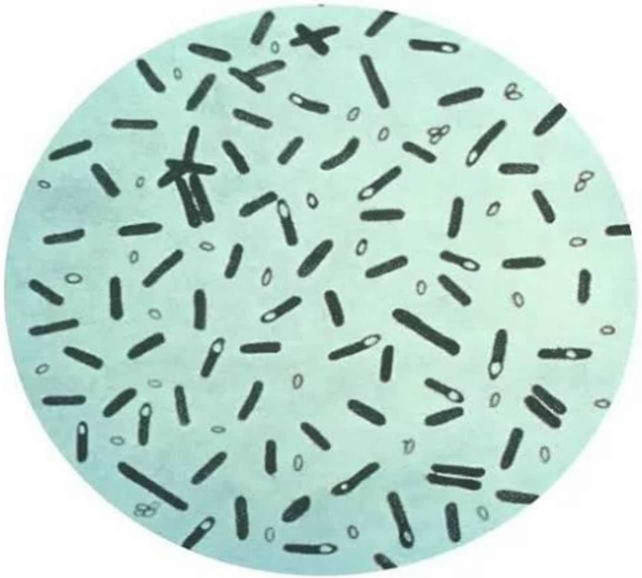

free endospore
vegetative cell
spore coat
mother
cell
developing
spore coat

The medical importance of spores:

1.Extraordinary resistance to heat and chemicals ⇨ sterilization cannot be achieved by boiling.

2.Sterilization by autoclaving at 121°C, usually for20 minutes.

3.Bacillus anthracis - spores used in biological warfare



Bacterial morphology

- They are prokaryotic cells.
- Bacterium is a single cell that can replicate by simple division called binary fission.

✓ **S : Stain.**
✓ **S :Shape.**
✓ **S :Size.**
✓ **S :Spore.**
✓ **M : Motility**
✓ **A :Arrangement.**
✓ **C :Capsule.**

SSSSMAC
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Classification of Microorganism

Acellular | Cellular

Cellular → Prokaryotes, Eukaryotes

Acellular

- They are not cells and have no cell membrane.
- They are composed of few genes protected by a protein coat

ex. Viruses

- They can live and reproduce only when inside a living cell.

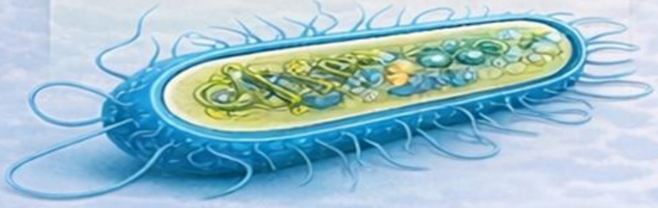
Prokaryotes

Pro → Before
Before nucleus, cells without nucleus

Have no nucleus

- Have no membrane-bound organelles
  less complex

Bacteria & Archeea

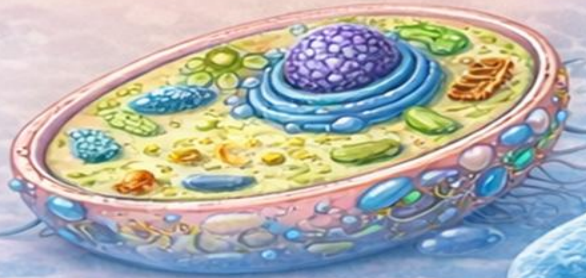


Eukaryotes

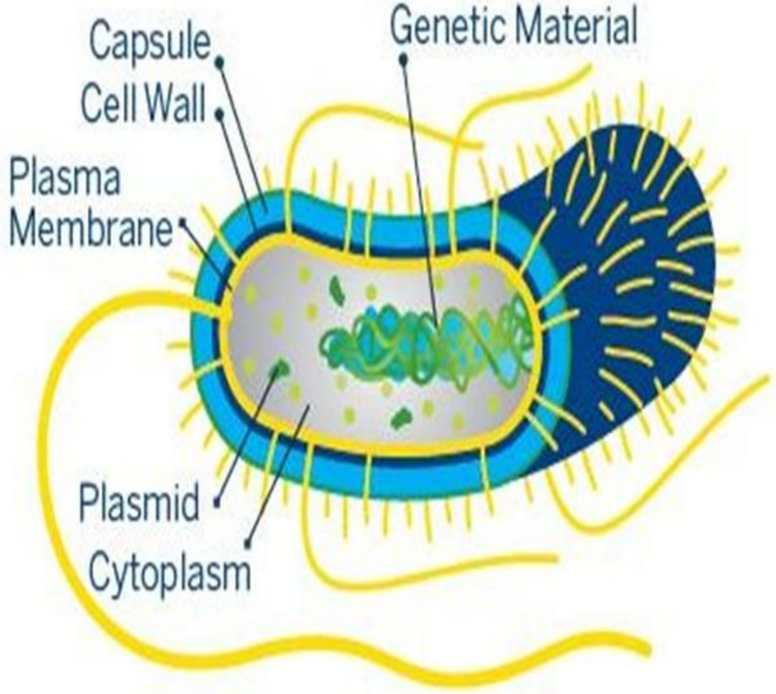
Eu → true or good
Cells that have nucleus

- Have true nucleus
- Many membrane-bond organelles

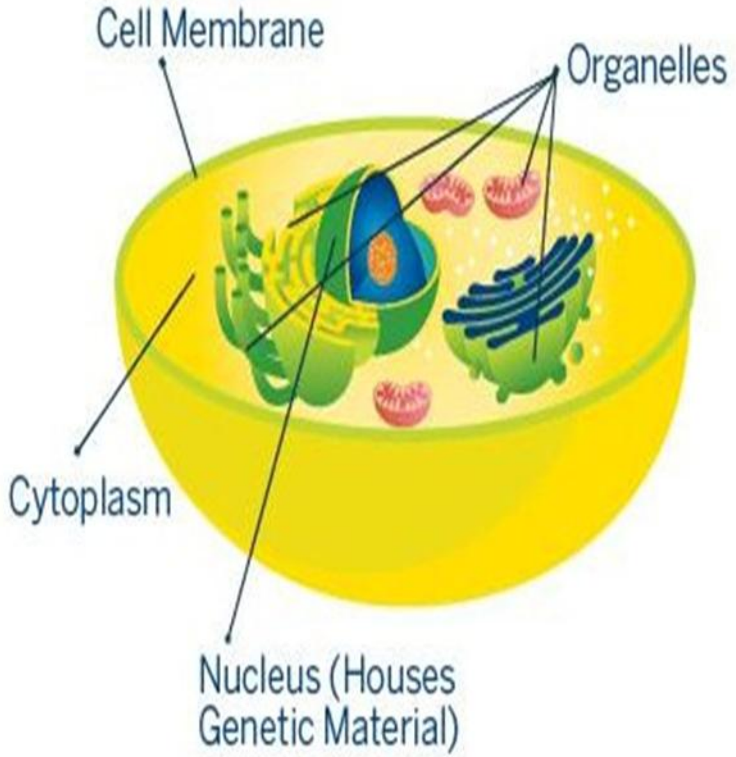
ex. Algae, protozoa, fungi, plants, animals, and humans.



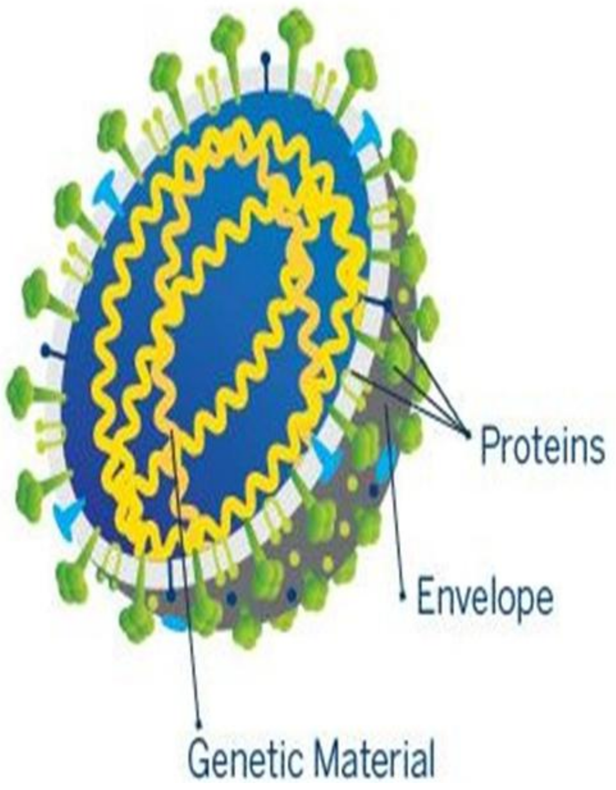
PROKARYOTE



EUKARYOTE



VIRUS



Characteristic	Eukaryotic Human Cells	Prokaryotic Bacterial Cells
Cell Wall containing peptidoglycan	No	Yes
Sterol in the Cytoplasmic membrane	Yes	No, except Mycoplasma
Cytoplasm	Organelles such as mitochondria and lysosomes: Yes	No
Size of ribosome	80S	70S
Nucleus	Present	Absent
Chromosome	More than one (linear)	Single (circular)


QUESTIONS

MCQ

1. Which of the following is NOT a bacterial shape?

A) Cocci
B) Bacilli
C) Spirilla
D) Cuboidal

2. Which is the most important differential stain in clinical microbiology?

A) Methylene blue
B) Crystal violet
C) Gram’s stain
D) Ziehl–Neelsen stain

3. Which component gives strength and rigidity to the bacterial cell wall?

A) Teichoic acid
B) Lipopolysaccharide (LPS)
C) Peptidoglycan
D) Ribosomes

4. Gram-positive bacteria are characterized by:

A) Thick peptidoglycan wall containing teichoic acids
B) Thin peptidoglycan wall and outer LPS layer
C) Absence of cell wall
D) Presence of mitochondria

5-A patient developed urinary tract infection caused by E.coli.
Which structure helped the bacteria to attach to the urinary tract?

a) Capsule
b) Flagella
c) Pili
d) Cytoplasm

Contacts

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Thank You

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